

# LEKHANA P

lekhanap2005@gmail.com | +91 9019648943 | Bengaluru

LeetCode | LinkedIn | GitHub

## EDUCATION

**JSS ACADEMY OF TECHNICAL EDUCATION**  
ELECTRONICS AND COMMUNICATION B.E  
CGPA: 8.3

Bengaluru, Karnataka  
2023 - 2027

**JAWAHAR NAVODAYA VIDYALAYA**  
PCMB (Class 12)-CBSE  
Percentage: 89.6%

Tumakuru, Karnataka  
2022 - 2023

**JAWAHAR NAVODAYA VIDYALAYA**  
(Class 10)CBSE  
Percentage: 93.8%

Tumakuru, Karnataka  
2020 - 2021

## SKILLS

Programming Languages: Java, C++, Python, html  
Libraries/Frameworks: Pandas, Scikit-learn, Matplotlib, TensorFlow, PyTorch  
Tools / Platforms: Git & GitHub, VS CODE, Jupyter Notebook, Google Colab  
Databases: SQL, MongoDB  
Technical Concepts: DSA, OOPS, DBMS, REST APIs  
AI / GenAI: LangChain, RAG (Retrieval-Augmented Generation), Google Generative AI, Vector Embeddings

## PROJECTS

### Legal Document Analyzer | Generative AI

November 2025-December 2025

FastAPI, LangChain, Google Generative AI, ChromaDB

- Developed a generative AI tool to demystify complex legal documents for a hackathon. Architected a Python backend using .
- FastAPI, orchestrating a Retrieval-Augmented Generation (RAG) pipeline with LangChain, Google's GenAI model, and a ChromaDB vector store.
- This AI-driven system delivered key features like automated summarization , risk analysis , and interactive Question and answer with source citations.

### Smart Specs | Assistive Wearable Device

August 2025-September 2025

Python, OpenCV, YOLOv8, BotSort, PyTorch, ESP32-CAM

- Developed a real-time monocular vision system for obstacle detection and spatial awareness using the pinhole camera model with focal length calibration and pixel-width-based depth approximation.
- Implemented multi-object tracking (BotSort) with a priority-based alert mechanism to maintain persistent object identities and announce only the nearest dynamic obstacle for efficient decision-making.
- Designed a modular, edge-ready architecture with scene-understanding triggers for contextual image description and future integration with embedded hardware (ESP32-CAM / edge AI devices).

### Tour Buddy | Travel Assistance App

June 2025-July 2025

Next.js, Node Js, Google Generative AI, MongoDB

- Architected and implemented an AI-powered travel planning platform leveraging server-side rendering and structured prompt workflows for personalized itinerary generation.
- Developed a modular, scalable frontend with schema-based validation and optimized state management to ensure reliability and cross-device responsiveness.
- Integrated geospatial visualization and asynchronous data-fetching strategies to enhance performance and enable dynamic, real-time travel insights.

## CERTIFICATIONS

- Oracle Cloud Infrastructure AI Foundations - **ORACLE**

## HONORS & AWARDS

- Awarded the prestigious **Reliance Undergraduate Scholarship**, for meritorious students across India for undergraduate studies.